THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** water	Splitting of water.
2	**(a)** receptor -> sensory neuron -> spinal cord -> motor neuron -> effector	Standard reflex pathway.
3	**(c)** Offspring produced by budding in Hydra	Asexual offspring are clones with minor copying errors.
4	**(b)** 1/2	Offspring Tt and tt in 1 : 1.
5	**(d)** herbivores such as the cow	Cellulose needs long digestion.
6	**(c)** auxin affects stem and root cells in opposite ways	Differential response.
7	**(c)** filament	The anther stalk.
8	**(c)** fertilisation	The sperm type decides.
9	**(d)** pulmonary artery	Only artery with deoxygenated blood.
10	**(d)** insulin	Diabetes.
11	**(b)** 23	Gametes have half the number.
12	**(a)** both alleles are recessive	Only tt shows dwarf.
13	**(d)** oxygen	Oxygen is a product.
14	**(d)** relay neuron	Relay neuron.
15	**(d)** Plasmodium	Multiple fission under unfavourable conditions.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	No light, no photosynthesis (1); Iodine test gives no blue-black colour (1).	1+1
17	Quick response avoids danger (1); Does not wait for brain processing (1).	1+1
18	Unicellular organisms are a single cell and can split (1); Complex organisms have organ systems and need specialised reproductive organs (1).	1+1
19	Dwarf offspring need a t allele from the tall parent (1); Tall parent is Tt (1).	1+1
20	Cellulose digestion takes longer (1); Meat is easier to digest (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Large flat surface to capture light (1); Stomata for gas exchange (1); Chloroplasts rich in chlorophyll in mesophyll (1).	1+1+1
22	Thinking and memory (1); Voluntary actions and speech (1); Senses like sight interpretation (1).	1+1+1
23	Spores are tiny and light, so they are dispersed far (1); Many do not find moisture and food (1); Large numbers ensure survival of some (1).	1+1+1
24	F1 Tt; F2 ratio 3 : 1 (1); Tall = 225, dwarf = 75 (1); Genotypes TT 75, Tt 150, tt 75 (1).	1+1+1
	OR: Mixing reduces the oxygen supplied to the body (1); Birds and mammals need more energy (1); Four chambers prevent mixing (1).	1+1+1
25	Low thyroxine causes pituitary to release more TSH (1); Thyroid makes thyroxine (1); Level returns to normal (1).	1+1+1
	OR: Male gametes are produced in large numbers (1); They are small and motile and many are lost before reaching the egg (1); Female gametes are few, larger and rich in food (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Destarch the plant in dark (1); Cover part of a leaf with black paper (1); Expose to sunlight (1); Iodine test: covered part no blue-black (1); Conclusion: light is necessary (1).	1+1+1+1+1

	OR: Receptor in skin (1); Sensory neuron to spinal cord (1); Relay and motor neuron (1); Effector muscle moves the foot (1); Brain informed later (1).	1+1+1+1+1
27	Binary fission in Amoeba and multiple fission in Plasmodium (1); Budding in yeast and Hydra (1); Fragmentation in Spirogyra and regeneration in Planaria (1); Spore formation in Rhizopus (1); Complex animals have specialised organs which cannot be rebuilt from fragments (1).	1+1+1+1+1
	OR: Gametes RY, Ry, rY, ry (1); 16 combinations (1); 9 round yellow (1); 3 round green and 3 wrinkled yellow (1); 1 wrinkled green (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** both Tt	3 : 1 ratio arises from Tt x Tt.
2	**(a)** cytoplasm	Cytoplasm.
3	**(d)** does not occur	Auxin source is lost.
4	**(b)** unisexual (male)	Only male parts are present.
5	**(a)** 1/2	Each birth is independent.
6	**(a)** 120/80 mm Hg	Standard.
7	**(a)** growth hormone	Pituitary.
8	**(b)** 46 chromosomes	Sperm (23) + egg (23).
9	**(a)** minor DNA copying errors	Copying errors create small differences.
10	**(c)** guard cells take up water and swell	Turgid guard cells open pore.
11	**(d)** muscle	Muscle contracts.
12	**(c)** Inaccurate DNA copying produces variation	Errors in copying provide the basis of variation.
13	**(a)** 40	1/4 of 160.
14	**(a)** much more energy per molecule of glucose	Complete breakdown in mitochondria gives more energy.
15	**(c)** abscisic acid	ABA.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Chemical is released only at axon terminal (1); Dendrite receives (1).	1+1
17	Only one parent copies its DNA (1); Only small copying errors occur, so offspring are almost identical (1).	1+1
18	Wrinkled (rr) are 1/4 (1); 100 seeds (1).	1+1
19	Carries digested fat from intestine (1); Drains excess fluid from tissues (1).	1+1
20	Regulate elongation, division (1); Coordinate responses (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Round yellow 9/16 = 450 (1); Round green 3/16 = 150, wrinkled yellow = 150 (1); Wrinkled green 1/16 = 50 (1).	1+1+1
22	Waste like urea accumulates (1); Dialysis removes waste (1); Kidney transplant is the long-term cure (1).	1+1+1
23	Hormones travel in blood (1); Need time to reach target (1); Chemical changes take time (1).	1+1+1
24	Quick multiplication of plants (1); Seedless plants like banana and orange can be grown (1); Desirable traits are maintained (1).	1+1+1
	OR: Blending would give intermediate seeds (1); Mendel found all F1 were round yellow (1); F2 showed original traits again, rejecting blending (1).	1+1+1
25	Mucus coats the wall (1); It protects against HCl (1); Acid helps pepsin work (1).	1+1+1
	OR: Impulse reaches the axon end (1); Chemical released into the synapse (1); Starts an impulse in the next dendrite (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Herbivores have long intestine (1); Carnivores short intestine (1); Cellulose digestion in herbivores (1); Caecum is large in herbivores (1); Carnivores digest meat easily (1).	1+1+1+1+1
	OR: Sugar rises after meal (1); Insulin released (1); Sugar stored/used (1); Diabetes: not enough insulin (1); Insulin injections treat (1).	1+1+1+1+1

27	Ovaries produce eggs and hormones (1); Oviduct is the site of fertilisation (1); Uterus receives and nourishes the embryo (1); Placenta exchanges materials with mother's blood (1); Cervix and vagina form the birth canal (1).	1+1+1+1+1
	OR: Genotypes TT, Tt and tt (1); Phenotypes tall and dwarf (1); T is dominant (1); t is recessive (1); T and t are alleles (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Life Processes, Control and Coordination, How do Organisms Reproduce?, Heredity** | Class 10 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 20 seeds	Each fertilised ovule forms one seed.
2	**(b)** Sperm decides the sex	The father's gamete is X or Y.
3	**(d)** xylem	Xylem.
4	**(c)** iodine	Goitre.
5	**(d)** umbilical cord	It carries nutrients to the foetus.
6	**(c)** an acquired trait due to environment	Environment affects the phenotype.
7	**(c)** no blue-black colour	No starch is made in dark.
8	**(c)** dendrite to axon end	Direction.
9	**(a)** vegetative propagation	Buds in the eyes sprout.
10	**(b)** TT	A homozygous genotype forms a single type of gamete.
11	**(c)** lactic acid accumulation	Anaerobic respiration.
12	**(d)** cytokinin	Cell division.
13	**(a)** self-pollination	Same plant, so genetically it counts as self-pollination.
14	**(a)** 23	22 + 1 = 23.
15	**(a)** transpiration pull	Pull on water column.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	A sticky surface holds the pollen (1); It increases the chance of pollen capture (1).	1+1
17	Round = 9 + 3 = 12 out of 16 (1); 3/4 (1).	1+1
18	More light energy is absorbed (1); Then other factors like CO_2 limit it (1).	1+1
19	Reflex: involuntary, spinal cord, fast (1); Voluntary: brain controls, slower (1).	1+1
20	Spores of Rhizopus fall on it and germinate (1); Hyphae spread rapidly and form many sporangia (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Vegetative propagation by suckers (1); Offspring have the same genes as the parent (1); Plants also bear fruits earlier (1).	1+1+1
22	Gametes R, r and R, r (1); Offspring RR, Rr, Rr, rr (1); Genotypic 1 : 2 : 1 and phenotypic 3 round : 1 wrinkled (1).	1+1+1
23	Heart valves prevent backflow (1); Vein valves keep blood moving towards heart (1); Blood flows one way (1).	1+1+1
24	Blood diverted to muscles (1); Heart rate increases (1); Skin blood vessels contract (1).	1+1+1
	OR: Sperms have to travel to reach the egg (1); Tail of the sperm helps swimming (1); Egg is large and non-motile, holding stored food (1).	1+1+1
25	Environment can change the phenotype (1); Plants: a genetically tall plant grows short in poor soil (1); Humans: tan or body weight depends on conditions (1).	1+1+1
	OR: Distances are long (1); Diffusion is slow (1); So respiratory pigments in blood are used (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Mouth: amylase gives sugar (1); Stomach: pepsin gives peptides (1); HCl provides acid (1); Small intestine: trypsin, lipase, bile (1); End products absorbed (1).	1+1+1+1+1
	OR: Cerebrum: thinking, reading (1); Cerebellum: riding a cycle (1); Medulla: salivation (1); Pons: breathing control relay (1); Midbrain: eye movements (1).	1+1+1+1+1

27	Bryophyllum leaf notches grow plantlets (1); Rose cutting grows roots and shoots (1); Potato eye sprouts into a new plant (1); All are vegetative propagation (1); New plants are genetically identical to the parent (1).	1+1+1+1+1
	OR: Parents TTRR x ttrr (1); F1 is TtRr, tall round (1); Tall round 90 (1); Tall wrinkled 30 and dwarf round 30 (1); Dwarf wrinkled 10 (1).	1+1+1+1+1

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